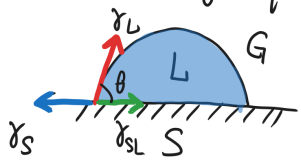
MATE664 L15

① The Young's equation & explanations (S: solid L: Liquid G: gas)



Remember γ : unit of $\begin{cases} N/m \Rightarrow \text{a force on contact line} \\ J/m^2 \end{cases}$

$$\gamma_S = \gamma_L \cos\theta + \gamma_{SL}$$

$$\cos\theta = \frac{\gamma_S - \gamma_{LS}}{\gamma_L} \quad \text{(Young's Equ.)}$$

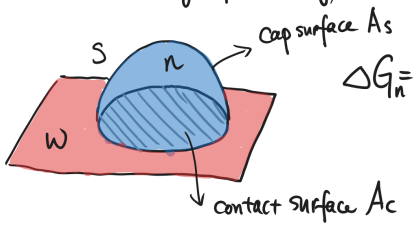
$$\cos\theta = \frac{\Delta W_{SL}}{\gamma_L} \quad \text{(Young-Dupré Equ)}$$

Also $\gamma_{SL} = \gamma_L + \gamma_S - \Delta W_{SL}$

$\gamma_S - \gamma_{SL} = \Delta W_{SL}$

Work of adhesion

② Estimating free energy barrier of nucleation in a cap



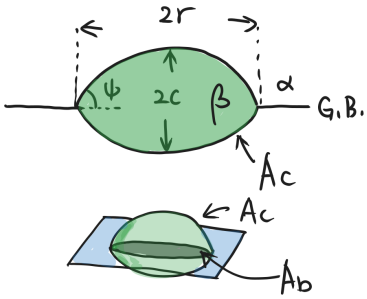
$$\Delta G_n = V_{cap} \cdot G^V + \gamma_{sn} A_s + (\gamma_{nw} - \gamma_{ws}) A_c$$

$$\frac{V_{cap}}{V_{sphere}} = \frac{2 - 3\cos\theta + \cos^3\theta}{4} = f$$

Interestingly we also have $\Delta G^{het} / \Delta G^{homo} = f$

Same results also apply for other geometries

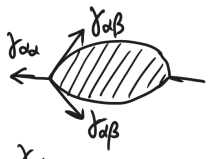
③ Lens-shape nucleation at $\alpha-\beta$ interface ?



$$\Delta G^{het} = V_{len} \cdot \Delta G^V + A_c \cdot \gamma_{\alpha\beta} + A_b (0 - \gamma_{\alpha\alpha})$$

$\gamma_{\alpha\alpha}$ is usually not 0 why? For G.B.

Young's equation for lens geometry



$$\gamma_{\alpha\alpha} = 2\gamma_{\alpha\beta}\cos\psi \quad \text{or} \quad \cos\psi = \frac{\gamma_{\alpha\alpha}}{2\gamma_{\alpha\beta}}$$

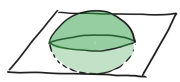
If $\gamma_{\alpha\alpha} = 0 \Rightarrow \cos\psi = 0 \Rightarrow \psi = \frac{\pi}{2} \Rightarrow$ $\boxed{\text{Nucleus is purely sphere!}}$

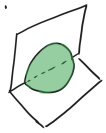
Still, we have $\dfrac{\Delta G^{het}}{\Delta G^{homo}} = 2f$

twice the volume of a cap

How to analyze different dimensionality ?

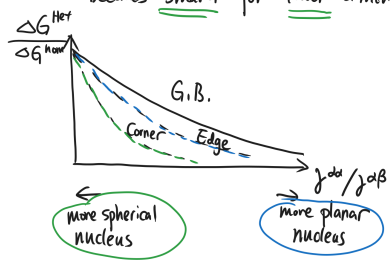
① Nucleation at G.B ② Nuleation at edge



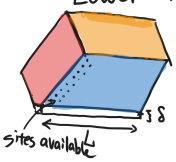


③ Nucleation at corner

As can be seen, the volume of nucleus becomes smaller for lower-dimensional nucleation



Lower nucleation barrier ≠ higher nucleation rate !



Consider the left geometry between 3 G.B.s

1. 2D / G.B. nucleation site density

$$n^B = \frac{N^{\#}\text{ sites}}{\text{Volume}} = \frac{n \cdot L^2 \cdot \delta}{L^3} = n\left(\frac{\delta}{L}\right)$$

(n: $N^{\#}$ of atoms/volume)

2. 1D / Edge

$$n^E = \frac{n \cdot L \cdot \delta^2}{L^3} = n \cdot \left(\frac{\delta}{L}\right)^2$$

3. 0D / corner

$$n^C = \frac{n \cdot \delta^3}{L^3} = n\left(\frac{\delta}{L}\right)^3$$

δ / L is usually small so $n^C \ll n^E \ll n^B \ll n$

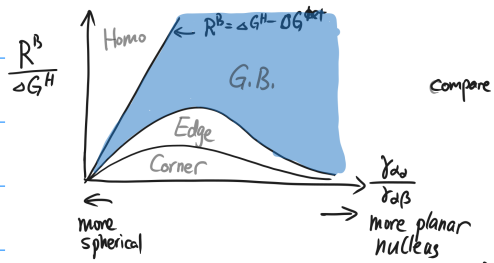
To compare J^{het} & J^{homo} (or J^B, J^E, J^C vs J^H)

$$J^{het,dim} = Z\beta_c \, e^{-\frac{\Delta G_c^{dim}}{k_B T}} \, n\left(\frac{\delta}{L}\right)^{3-dim}$$

$$J^{het} / J^H = \left(\frac{\delta}{L}\right)^{3-dim} e^{-\frac{\Delta G_c^{dim} - \Delta G^H}{k_B T}}$$

or $$\ln\left(\frac{J^{het}}{J^H}\right) = \underbrace{(3-dim)\left(\frac{\delta}{L}\right)}_{\text{Dimensionality term}} + \underbrace{\left(\frac{\Delta G_c^H - \Delta G_c^{het}}{k_B T}\right)}_{\text{Barrier term}}$$

From the nucleation rate chart $R^B = k_B T \ln\left(\frac{L}{\delta}\right)$ for G.B



compare barrier

Edge & corner nucleation rates are faster when $R^B / \Delta G^H$ is small

GB rate is low, But power (3-dim) enhances

Lecture 1

Molecular Interactions at Interfaces

The concept of **interface** refers to the boundary that separates two regions (phases) with different physical properties. Historically, when one of the phase is air / vacuum, people also refer such interface simply as **surface**. Physical and chemical phenomena associated with the interfaces exist at every length scale in our daily life, from proteins in fluid to macroscopic to circulation in atmosphere. In this lecture, we will briefly discuss about the molecular interactions at the interfaces between different materials. We will show a quantitative description of the surface tension based on molecular interactions.

1.1 The surface tension

The key property of an interface is known as the **surface tension** (or **interfacial tension** when both phases are liquid / solid), which is associated with the energy required to form the interface.[1] Here we will briefly show the origin of the surface tension and how it is related to the molecular interactions at the interface.

Consider a homogeneous bulk material (can be either liquid / solid) consists of molecule A (Figure 1.1).

The bulk cohesive energy $E_{\mathrm{A,bulk}}$ per mole is calculated by counting the pairwise interaction energy w_{AA} between all the neighboring atoms, such that:

$$E_{\mathrm{A,bulk}} = \frac{1}{2} w_{\mathrm{AA}} N_{\mathrm{A}} Z_{\mathrm{b}} \tag{1.1}$$

where $w_{\mathrm{AA}} < 0$ is the pairwise interaction energy between two A atoms at the equilibrium distance, N_{a} is the Avogadro constant, and Z_{b} is average number of neighboring atoms. We will discuss about the meaning of the "equilibrium distance" later. The pre-factor 1/2 is added to avoid doubling counting.

Let's now look at the quantity w_{AA}. From the Lennard-Jones (LJ) model[2] of intermolecular forces, w_{AA} is negative at the equilibrium inter-molecular distance r_0 (see Figure 1.2), where r_0 is defined as the distance that $\mathrm{d}w_{\mathrm{AA}}/\mathrm{d}r = 0$. As a result, we have $E_{\mathrm{A,bulk}} < 0$, indicating that formation of bulk material is a energetically favored process. In other words, we need to pay energy to separate the bulk material and create interfaces. For the sake of

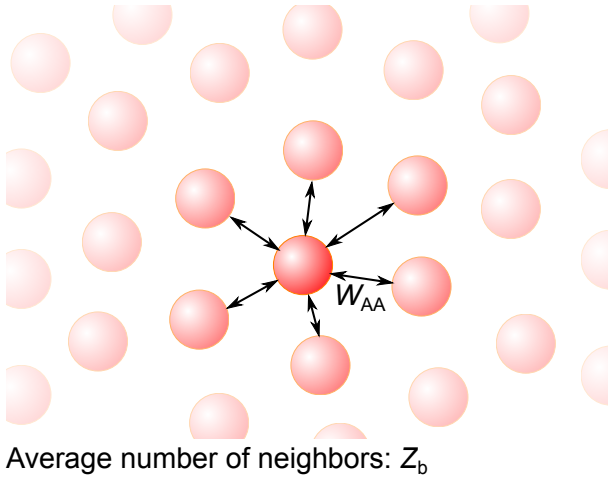


Figure 1.1: Scheme of a homogeneous bulk material. The cohesive energy $E_{A,bulk}$ is calculated from all the pairwise interaction energy.

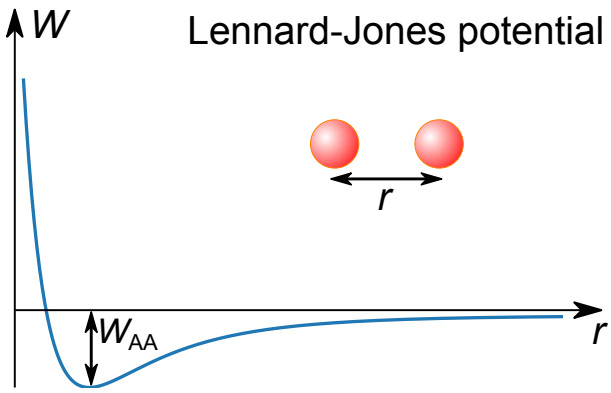


Figure 1.2: Scheme of the Lennard-Jones model of intermolecular interaction. w_{AA} is negative at the equilibrium inter-molecular distance r_0

comparison, we also define the bulk cohesive energy *per molecule*, $\tilde{E}_{\mathrm{A,bulk}} = E_{\mathrm{A,bulk}}/N_{\mathrm{A}}$. Consider an interface between material A and vacuum (Figure 1.3), we can see that the average number of neighboring atoms on the surface becomes Z_{s} (smaller than Z_{b}), and similarly, we can define the surface cohesive energy per molecule as:

$$\tilde{E}_{\mathrm{A,surf}} = \frac{1}{2} w_{\mathrm{AA}} Z_{\mathrm{s}} \tag{1.2}$$

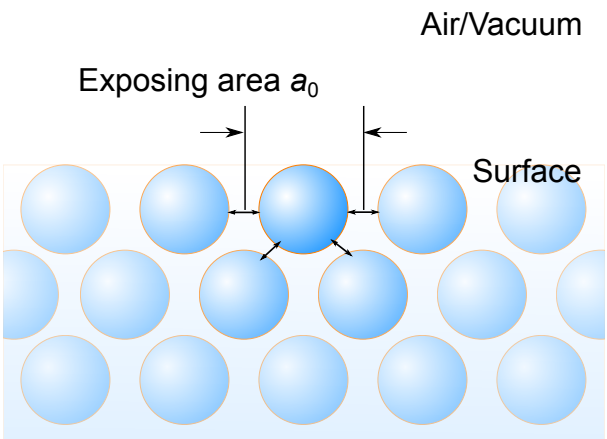


Figure 1.3: Scheme of the molecules on the surface. The coordinate number Z_{s} is smaller than that in the bulk material.

Obviously, we have $\tilde{E}_{\mathrm{A,surf}}$ less negative than $\tilde{E}_{\mathrm{A,bulk}}$ due to $Z_{\mathrm{s}} < Z_{\mathrm{b}}$. We define the **surface tension** γ as the energy required to create the interface per unit area. Consider that each molecule A occupies a surface area of a_0. The surface tension between A and vacuum / air, γ_{A}, can be written using our previous knowledge:

$$\begin{aligned} \gamma_{\mathrm{A}} &= \frac{\tilde{E}_{\mathrm{A,surf}} - \tilde{E}_{\mathrm{A,bulk}}}{a_0} \\ &= \frac{1}{2a_0} w_{\mathrm{AA}} (Z_{\mathrm{s}} - Z_{\mathrm{b}}) \end{aligned} \tag{1.3}$$

which is **always** larger than 0. Note that this conclusion is only valid when discussing the interface with vacuum / air, as we will see the later the definition of interfacial tension γ_{AB} is different.

The unit of surface tension is [Energy]/[Area], usually expressed in J·m^{-2} or mJ·m^{-2}. Alternatively the surface tension is also written in unit of [Force]/[Length], e.g. N·m^{-1}, which characterizes the force acting per length acting along the interfacial boundary line (the origin of the term "tension"). The surface tension of different materials can vary quite a lot, here are the values of surface tension for several typical liquids:[3]

- γ_{Ag} (1200 °C) = 878 mJ·m^{-2}
- γ_{Hg} (25 °C) = 489 mJ·m^{-2}

- γ_{Water} (25 °C) = 72.8 mJ·m^{-2}
- γ_{Octane} (25 °C) = 20 mJ·m^{-2}

Such difference in the value of γ has great impact on some physical properties of the liquids. Here are several questions that you can think of:

1. Is it easier for a solid object to float on water (γ = 72.8 mJ·m^{-2}) or ethanol (γ = 22 mJ·m^{-2})?
2. Is it easier to create bubbles from pure water or water with 1 mM surfactant ($\gamma \approx$30 mJ·m^{-2})?

1.2 The interfacial tension

Similar to the case of a surface in vacuum / air, there is also an interfacial tension between two different materials. For a interface formed between two phases consist of molecules A and B, respectively (Figure 1.4), the interfacial tension γ_{AB} is defined as the energy per area required to create such interface. We can calculate γ_{AB} in the following way:

- Separate the interface to get the surfaces of materials A and B, respectively. This process requires ΔW_{AB} (work of adhesion) per unit area.
- The two newly created surfaces of A and B have surface energies of γ_{A} and γ_{B}, respectively.

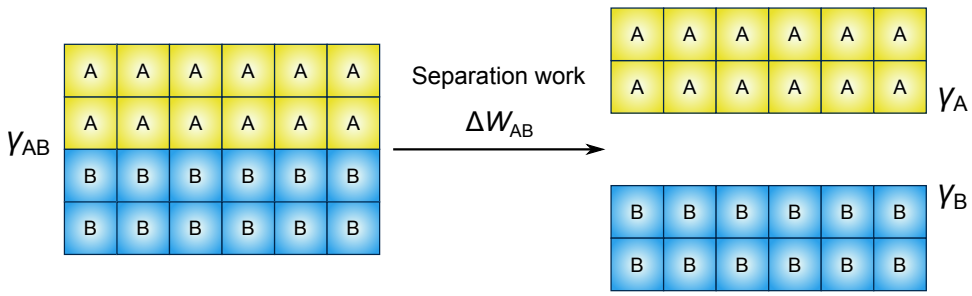


Figure 1.4: The interfacial tension between A and B phases. A work of adhesion ΔW_{AB} is required to separate the interface and create new surfaces of A and B with vacuum / air.

In total, the interfacial tension between A and B is:

$$\gamma_{\text{AB}} = \gamma_{\text{A}} + \gamma_{\text{B}} - \Delta W_{\text{AB}} \tag{1.4}$$

A very crude assumption is that the coordinate numbers Z_{s} and Z_{b}, as well as the average area a_0 are the same for both A and B phases. Under such consideration we can express γ_{AB} as:

$$\gamma_{\text{AB}} = \frac{1}{a_0}(Z_{\text{b}} - Z_{\text{s}})\left[w_{\text{AB}} - \frac{1}{2}(w_{\text{AA}} + w_{\text{BB}})\right] \tag{1.5}$$

where w_{AA}, w_{BB} and w_{AB} are the intermolecular energies between A-A, B-B and A-B, respectively. Now we see that γ_{AB} depends on the relative magnitudes between 3 interaction energies. As we will see later, this explains the different wetting behaviors of surfaces.

However, in real materials, the assumption of same Z_b and Z_s may not apply. According to the Lennard-Jones model we discuss in Figure 1.2, the intermolecular potential between two molecules A and B, $\Phi(r)$, as a function of the intermolecular distance r, consists of an attraction term ($\propto -1/r^6$) and a repulsive term ($\propto 1/r^{12}$). An example of the LJ potential is shown in Figure 1.5.

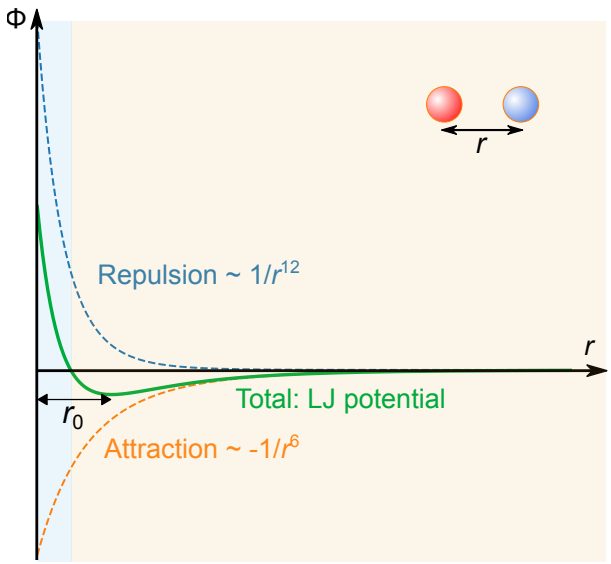


Figure 1.5: Lennard-Jones potential $\Phi(r)$ between two molecules as a function of r (green solid line), constructed by a repulsive term (blue dashed line) and an attraction term (red dashed line).

Since the repulsive part is prominent only for distances close to the inner core of the atoms, we can consider the $1/r^6$ attractive potential energy alone, i.e. $\Phi(r) = -\beta/r^6$, where $\beta\,(>0)$ is the coefficient associated with the van der Waals (vdW) interaction. At equilibrium distance r_0, $\Phi(r_0)$ becomes the w_{AB} that we define previously in Equation 1.5.

Consider two semi-infinite objects 1 (consists of molecule A) and 2 (consists of molecule B) with separation d as shown in Figure 1.6, the total vdW potential energy between the two objects Φ_{12}, is calculated as the summation of pairwise interaction potential energy, such that:

$$\begin{aligned}\Phi_{12}(d) &= \sum_{N_1}\sum_{N_2} -\frac{dN_1 dN_2}{2}\frac{\beta}{r^6} \\ &= \sum_{dV_1}\sum_{dV_2} -\frac{(\rho_1 dV_1)(\rho_2 dV_2)}{2}\frac{\beta}{r^6} \\ &= -\frac{\beta\rho_1\rho_2}{2}\iint_{1,2}\frac{dV_1 dV_2}{r^6}\end{aligned} \tag{1.6}$$

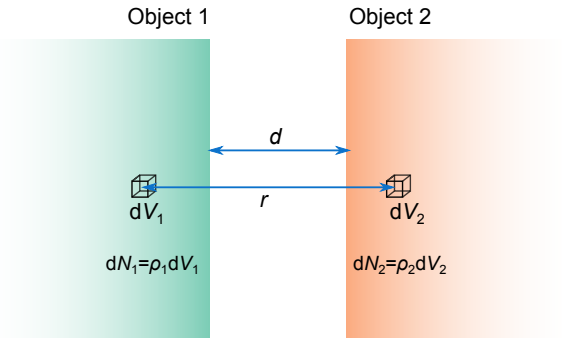


Figure 1.6: Scheme for the summation of pairwise interaction potential between two macroscopic objects 1 and 2.

where ρ_1, ρ_2 are the number densities of molecules in objects 1 and 2, respectively, $\mathrm{d}V_1$ and $\mathrm{d}V_2$ are infinitely small volumes in 1 and 2, respectively, and r is the distance between $\mathrm{d}V_1$ and $\mathrm{d}V_2$. The integral is done with $r = d$ to $r = \infty$.

We can work out the integral in Equation 1.6 in two steps: (i) get the potential energy between a single molecule A with the bulk object 2 $\Phi_{\mathrm{A}\to 2}$, and (ii) integrate $\Phi_{\mathrm{A}\to 2}$ inside the object 1. For $\Phi_{\mathrm{A}\to 2}$, the integral of $\mathrm{d}V_2$ can be efficiently carried out using the ring integral (Figure 1.7):

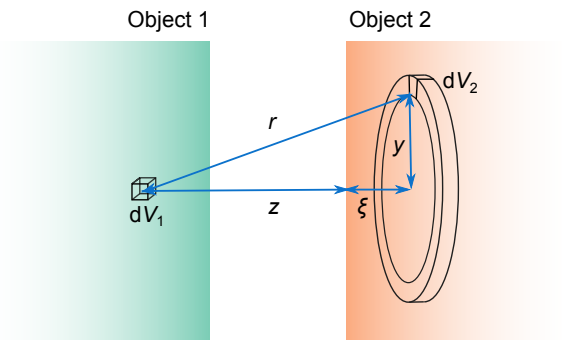


Figure 1.7: Ring integral of the interaction between a molecule A and the macroscopic object 2.

$$\begin{aligned}\Phi_{\mathrm{A}\to 2} &= -\int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} \mathrm{d}\xi \int_0^{\infty} \mathrm{d}y \frac{\beta\rho_2 y}{(\sqrt{(z+\xi)^2+y^2})^6} \\ &= -\frac{\pi\beta\rho_2}{6z^3}\end{aligned} \tag{1.7}$$

where z is the distance from A to the interface of object 2, ξ, y and θ are the local coordinates inside object B. We can see from Equation 1.7, that the power law of the vdW interaction changes from $1/r^6$ to $1/z^3$, a consequence of the existence of all pairwise interactions. The rest of our task is to further integrate $\Phi_{\mathrm{A}\to 2}$ inside object 1. Consider the area of the object

1 is S, we have

$$\begin{aligned}\Phi_{12}(d) &= \int_d^\infty \Phi_{\mathrm{A}\to 2} S\rho_1 \mathrm{d}z \\ &= S\int_{\mathrm{d}}^\infty -\frac{\pi\beta\rho_1\rho_2}{6z^3}\mathrm{d}z \\ &= -\frac{\pi\beta\rho_1\rho_2}{12d^2}S\end{aligned} \tag{1.8}$$

Consider the fact that $\Phi_{12}(d \to \infty) = 0$, we can relate ΔW_{AB} with Φ_{12} by $\Delta W_{\mathrm{AB}}(d) = [\Phi_{12}(d \to \infty) - \Phi_{12}(d)]/S = -\Phi_{12}(d)/S$, which is finally written as:

$$\begin{aligned}\Delta W_{\mathrm{AB}} &= \frac{\pi\beta\rho_1\rho_2}{12d^2} \\ &= \frac{A_{12}}{12\pi d^2}\end{aligned} \tag{1.9}$$

where the constant $A_{12} = \pi^2\beta\rho_1\rho_2$ is called the Hamaker constant[4] between objects 1 and 2. The Hamaker constants for different materials can be found in the literature (for example Reference [5]) , making it the one of the standard parameters in surface science. We see again that the power law of the interaction energy between two bulk objects further changes to $1/d^2$, i.e. the range vdW interaction between two macroscopic objects is much longer than intermolecular interactions.

With this knowledge we can finally calculate γ_{A}, γ_{B} and γ_{AB}. We can easily find that $\gamma_A = -\Delta W_{\mathrm{AA}}/2$ due to its definition, therefore we get: $\gamma_A = \dfrac{A_{\mathrm{AA}}}{24\pi d_{\mathrm{AA}}^2}$ and similar for γ_B. For γ_{AB} we have:

$$\gamma_{\mathrm{AB}} = \gamma_{\mathrm{A}} + \gamma_{\mathrm{B}} - \Delta W_{\mathrm{AB}} = \frac{A_{\mathrm{AA}}}{24\pi d_{\mathrm{AA}}^2} + \frac{A_{\mathrm{BB}}}{24\pi d_{\mathrm{BB}}^2} - \frac{A_{\mathrm{AB}}}{12\pi d_{\mathrm{AB}}^2} \tag{1.10}$$

Usually we assume that $d_{\mathrm{AA}} \approx d_{\mathrm{BB}} \approx d_{\mathrm{AB}}$ (see Reference [6], Chapter 13.13), and $A_{\mathrm{AB}} \approx \sqrt{A_{\mathrm{AA}}A_{\mathrm{BB}}}$, as a consequence we can estimate γ_{AB} using:

$$\gamma_{\mathrm{AB}} \approx \left(\sqrt{\gamma_{\mathrm{A}}} - \sqrt{\gamma_{\mathrm{B}}}\right)^2 \tag{1.11}$$

which indicates γ_{AB} is usually positive. There are several indications from Equation 1.11:

- The interfacial tension between the same material γ_{AA} is always 0.
- The high energy surface dominates the interfacial tension, i.e. $\gamma_{\mathrm{AB}} \approx \gamma_{\mathrm{A}}$ when $\gamma_{\mathrm{A}} \gg \gamma_{\mathrm{B}}$, and *vice versa*.

So far we have see how the surface and interfacial tensions come from molecular interactions with an intuitive theory. We will discuss more about the origin of the parameter β and Hamaker constant A of the vdW interactions as well as the modern understanding of the interfacial interactions in Lectures 9 and 10.

References

(1) In *Adhesives Technology Handbook (Second Edition)*, Ebnesajjad, S., Ed.; William Andrew Publishing: Norwich, NY, 2009, pp 21–36.

(2) Jones, J. E. *Proceed. R. Soc. A* **1924**, *106*, 463–477.

(3) Adamson, A. W.; Gast, A. P., *Physical chemistry of surfaces*, 6th ed; Wiley: New York, 1997.

(4) Hamaker, H. *Physica* **1937**, *4*, 1058–1072.

(5) Bergström, L. *Adv. Colloid Interface Sci.* **1997**, *70*, 125–169.

(6) Israelachvili, J. N., *Intermolecular and surface forces*, Third edition; Elsevier, Academic Press: Amsterdam, 2011.

Lecture 2

Wetting and Spreading

In Lecture 1, we have discussed the concept of surface tension. Increasing the surface area always raises the free energy of the system. As a consequence there is a tendency to reduce surface area. A well-known example of this phenomenon is that a suspended water droplet is always spherical when gravity is negligible (i.e. in space station or at microscopic scale). However, the situation becomes different when we consider the interface between solid / liquid phase A with a flat surface and liquid phase B. When the two phases contact, the liquid B may either (i) spreads on A, forming a thin liquid film, or (ii) partially wet B, forming a droplet (Figure 2.1).

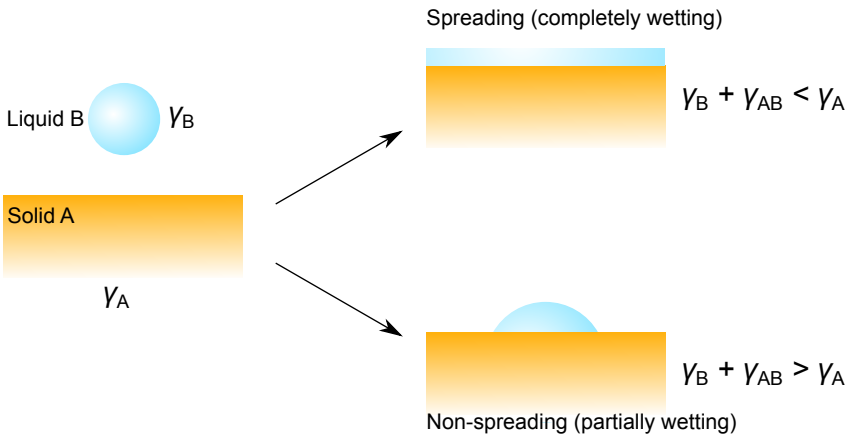


Figure 2.1: Spreading and partial wetting at the liquid-solid interface.

What factors determine the formation of a wetting later / droplet? Can we model the shape of the droplet? In this lecture we will try to answer the above question using our knowledge of surface tension.

2.1 The relation between wetting and surface tension

The criteria determining whether spreading or partial wetting happens, is the relative magnitudes between γ_A, γ_B and γ_{AB}. We can define a quantity "spreading coefficient",[1] $S_{B/A}$,

such that:

$$S_{B/A} = \gamma_A - (\gamma_B + \gamma_{AB}) \tag{2.1}$$

If $S_{B/A} > 0$, liquid A can complete wet B, and *vice versa*. This can happen when γ_A is large. As an example we can use the spreading coefficient to explain the wetting between different liquids. As shown in Table 2.1, low surface tension liquids (hexane, benzene and oleic acid) completely wet on the water surface and form films, while CS_2 and CH_2I_2 do not wet on water and form droplets. On the contrary, all these liquid completely wet on the surface of mercury, due to the extremely large γ. However, we should see that the spreading behavior cannot be simply judged from γ_A or γ_B alone: oleic acid and CS_2 have almost the same surface tension, however oleic acid spreads on water while CS_2 only partially wet. In other words, our simplified model for the interfacial tension γ_{AB} from Equation 1.11 is not sufficient to explain some real liquids.

Table 2.1: Spreading coefficient $S_{B/A}$ for different liquids on water and mercury surface. All data are in mJ·m^{-2}.

S \ A / B	Water γ=72	Mercury γ=489
Hexane $\gamma \approx$20	3.4	79
Benzene $\gamma \approx$30	8.9	99
Oleic acid $\gamma \approx$35	24.6	102
CS_2 $\gamma \approx$35	-8.2	102
CH_2I_2 $\gamma \approx$50	-26.5	70

2.2 The contact angle

Although the spreading coefficient $S_{B/A}$ is useful to distinguish whether a liquid spreads on the surface, in the case of partial wetting, we still want to quantify how "well" the liquid wets. We can do this by measure the **contact angle** of a liquid droplet (L) on a solid surface (S) in air (Figure 2.2). The contact angle is the angle between the liquid-air and liquid-

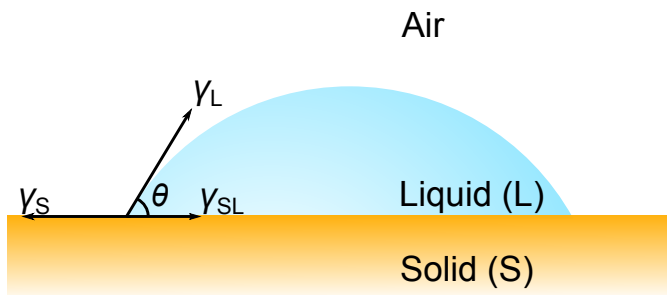


Figure 2.2: The relationship between the contact angle of a droplet θ, and the surface tensions.

solid contact lines, and is highly related with the surface tensions. Recall in Lecture 1 the surface tension also have unit of [force]/[length], in fact, in a microscopical view, the surface tension is a force that operates along the interface contact line,[2] which tends to decrease the surface area. Near the triple-phase contact line of a droplet at equilibrium, there is a force balance between the surface tensions γ_L, γ_S and γ_{SL}:

$$\gamma_S = \gamma_L \cos\theta + \gamma_{SL} \tag{2.2}$$

which is known as the Young equation[3] of contact angle. Use the relation in Equation 1.4, the Young equation can also be rewritten as:

$$\gamma_L(1 + \cos\theta_Y) = \Delta W_{SL} \tag{2.3}$$

which is also known as the Young-Dupré equation,[4] where θ_Y is the static contact angle described by the Young-Dupré equation. Equation 2.3 relates two macroscopic (and easily measurable) quantities θ_Y and γ_L with the work of adhesion ΔW_{AB}. From our microscopic model in Lecture 1, θ_Y can be written using the Hamaker constant A_{SL} and equilibrium solid-liquid interface distance d_{SL} as:

$$\theta_Y = \arccos\left[\frac{A_{SL}}{12\pi d_{SL}^2}\frac{1}{\gamma_L} - 1\right] \tag{2.4}$$

Since d_{SL} for almost all liquid-solid interface is usually at the order of 5 Å, the contact angle can be qualitatively inferred from A_{SL} and γ_L:

- Higher A_{SL}: the adhesion between S and L is strong → smaller θ_Y (more wetting)
- Higher γ_L: more energy is required for wetting → larger θ_Y (less wetting)

An extreme case is when $\Delta W_{SL} = 0$, we have θ_Y=180°, i.e. the liquid cannot wet the surface at all.

2.3 Measuring surface tension

As you can imagine, the surface tension of a liquid can be measured as a force acting on the boundary, using an instrument called the **tensiometer**. A widely-used design of the tensiometer is the Wilhelmy plate method, named after its inventor Ludwig Wilhelmy. A typical setup for the Wilhelmy plate can be seen in Figure 2.3, where a thin plate (S) with thickness d and width W ($d \ll W$) is immersed in a liquid (L) with an immersion depth of H. The liquid forms contact angle θ on both sides of the plate. When moving the plate in the z direction, a total force F_{tot} is measured using a micro balance, which is further converted to the surface tension γ.

The total force $F_{tot} = F_{wetting} + F_{bouy} + F_{weight}$, where $F_{wetting}$ is the wetting force on the contact line, $F_{bouy} = -\rho_L g d W H$ is the buoyancy force, and F_{weight} is the gravity of the plate.

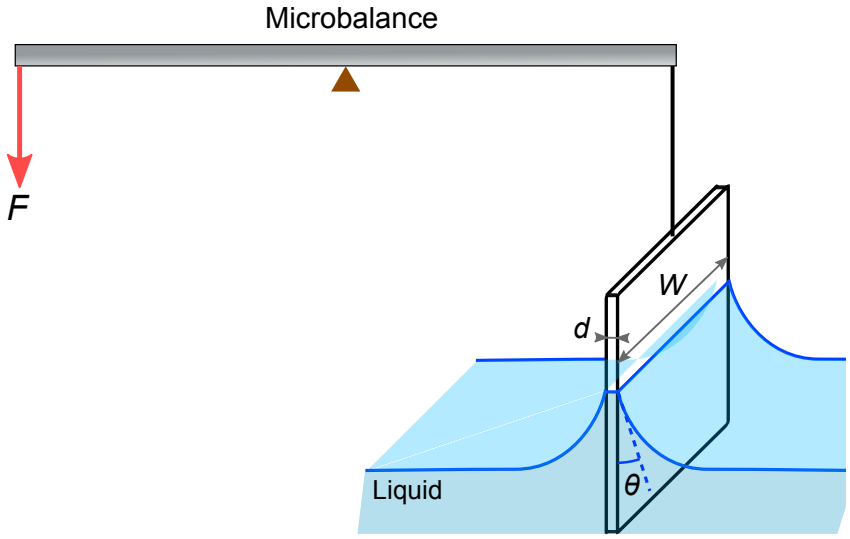


Figure 2.3: Scheme of the Wilhelmy plate used to measure the surface tension of a liquid.

Both F_{bouy} and F_{weight} can be determined using geometric parameters and are usually very small compared with F_{wetting} since $d \ll W$. At steady state, F_{wetting} is expressed as:

$$F_{\text{wetting}} = \gamma_{\text{L}} \cos\theta[2(d + W)] \approx 2\cos\theta\gamma_{\text{L}}W \tag{2.5}$$

If $\theta \to 0$, i.e. complete wetting on the plate, we can estimate the surface tension as $\gamma_{\text{L}} = F_{\text{wetting}}/2W \approx F_{\text{tot}}/2W$. However this equation does not work for larger contact angles, and should used with care.

Besides the Wilhelmy plate method, there are many other techniques to determine the surface tension γ, including the capillary rise method,[5] pendant droplet method,[6] Du Nouy ring method,[7] etc. Each method has its advantages and limitations, and has to be chosen with care.

2.4 Wetting on real surfaces

The discussion about the contact angle so far has been made regarding the flat surface. A water contact angle larger than 120° is quite unusual for flat surfaces.[8] This is not hard to imagine, using the simple model that $\gamma_{\text{SL}} = (\sqrt{\gamma_{\text{S}}} - \sqrt{\gamma_{\text{L}}})^2$, we get $\Delta W_{\text{SL}} = 2\sqrt{\gamma_{\text{S}}\gamma_{\text{L}}}$. Plug it into Equation 2.3, we have:

$$\cos\theta_{\text{Y}} = 2\sqrt{\frac{\gamma_{\text{S}}}{\gamma_{\text{L}}}} - 1 \tag{2.6}$$

If θ_{Y} is 120°, we need γ_{S} as low as 5 mJ·m^{-2}, which is even much smaller than teflon. Although such analysis oversimplifies γ_{SL}, we can infer that a higher θ_{Y} than 120° for water on a ideally flat surface, is merely unphysical.

However, in daily life we often observe the contact angle to be much larger than 150° on some plant leaves and flower pedals which have rough surfaces. Usually on a rough

surface, the macroscopic static contact angle $\theta^\star$, is different from the static contact angle θ_Y on a flat surface. Depending on the relative magnitudes between $\theta^\star$ and θ_Y, there are two models to describe the wetting on a rough surface (Figure 2.4)

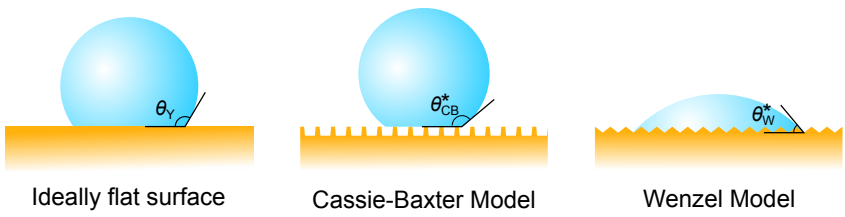


Figure 2.4: Wetting on a flat surface and on a rough surface described by Cassie-Baxter and Wenzel models.

- $\theta^\star > \theta_Y$ (Cassie-Baxter model[9])

  In the Cassie-Baxter (CB) model, there are air voids within the rough structure on the surface that cannot be wetted, and therefore surface roughness leads to less wetting. $\theta^\star$ in the CB model is written as:

$$\cos\theta^\star_{CB} = rf\cos\theta_Y + f - 1 \tag{2.7}$$

  where r is the roughness factor of the surface, and f is the fraction of wetted surface area

- $\theta^\star < \theta_Y$ (Wenzel model[10])

  In the Wenzel (W) model, the liquid can completely wet the rough structure, and leads to more wetting. The fraction f in the CB model becomes 1 and $\theta^\star$ is simply controlled by r

$$\cos\theta^\star_W = r\cos\theta_Y \tag{2.8}$$

In the end, we can see that both models have a similar physical background, and the Wenzel model is the special case of Cassie-Baxter model when $f = 1$. The different wetting phenomena lead to some interesting properties. Usually for a **hydrophobic** material with (θ_Y >90°), surface roughness leads to the Cassie-Baxter wetting state, and $\theta^\star$ can approach 180° (**superhydrophobic**). On the other hand, for a **hydrophilic** material (θ ≪90°), the surface roughness leads to the Wenzel wetting states and makes the surface **superhydrophilic** ($\theta^\star \to 0°$).

On a real surface, the contact angle of a moving contact line can be different from the static contact angle $\theta^\star$. The contact angle when the contact line advances is called the **advancing angle** θ_{adv}, and the contact angle when the contact line shrinks is called the **receding angle** θ_{rec} (Figure 2.5). The difference between the dynamic contact angles comes from the local defects that "pins" the contact line, and we have $\theta_{adv} > \theta^\star > \theta_{rec}$. This phenomenon is known as the **contact angle hysteresis** and several models have been proposed to explain its origin.[11–13]

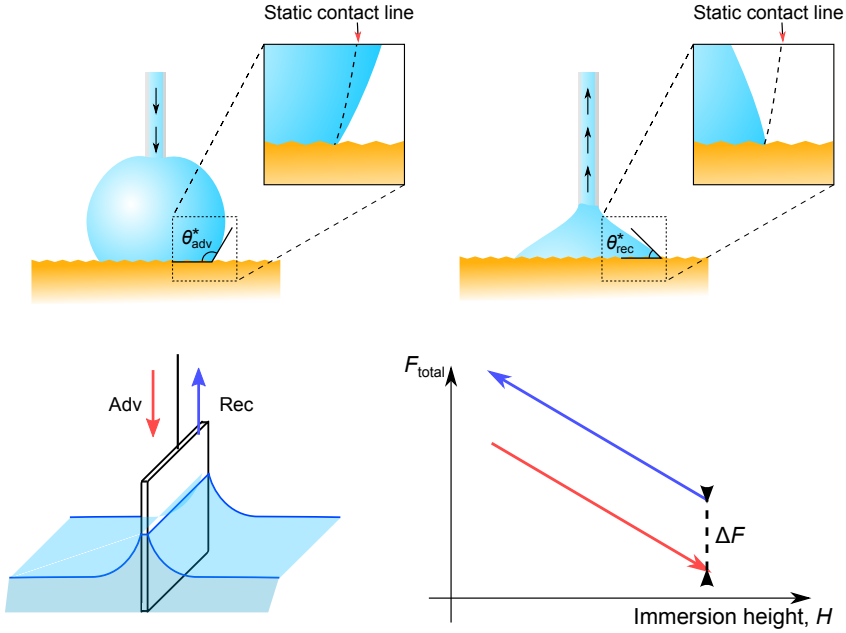


Figure 2.5: Dynamic contact angles and the measurement of contact angle hysteresis from Wilhelmy plate method.

An ideally flat surface should have contact angle hysteresis close to 0°. The hysteresis, defined as the difference between θ_{adv} and θ_{rec} is also measurable through the Wilhelmy plate method. As shown in Figure 2.5, the force curve of total force F_{tot} as a function of immersion height h do not overlap between the advancing (dipping) and receding (lifting) processes. The displacement force ΔF between the two curves can be written using our previous knowledge as:

$$\Delta F \approx 2W\gamma_{\mathrm{L}}(\cos\theta_{\mathrm{rec}} - \cos\theta_{\mathrm{adv}}) \tag{2.9}$$

References

(1) Ross, S.; Becher, P. *J. Colloid Interface Sci.* **1992**, *149*, 575–579.

(2) Marchand, A.; Weijs, J. H.; Snoeijer, J. H.; Andreotti, B. *Am. J. Phys.* **2011**, *79*, 999–1008.

(3) Young, T. *Phil. Trans. R. Soc. of Lond.* **1805**, *95*, 65–87.

(4) Schrader, M. E. *Langmuir* **1995**, *11*, 3585–3589.

(5) Richards, T. W.; Carver, E. K. *J. Am. Chem. Soc.* **1921**, *43*, 827–847.

(6) Berry, J. D.; Neeson, M. J.; Dagastine, R. R.; Chan, D. Y.; Tabor, R. F. *J. Colloid Interface Sc.* **2015**, *454*, 226–237.

(7) Du Nouy, P. L. *J. Gen. Physiol.* **1925**, *7*, 625–631.

(8) Quéré, D. *Physica A* **2002**, *313*, 32–46.

(9) Cassie, A. B. D.; Baxter, S. *Trans. Faraday Soc.* **1944**, *40*, 546.

(10) Wenzel, R. N. *Ind. Eng. Chem.* **1936**, *28*, 988–994.

(11) Joanny, J. F.; de Gennes, P. G. *J. Chem. Phys.* **1984**, *81*, 552–562.

(12) Patankar, N. A. *Langmuir* **2010**, *26*, 7498–7503.

(13) Yeh, K.-Y.; Chen, L.-J.; Chang, J.-Y. *Langmuir* **2008**, *24*, 245–251.

Lecture 5

Supercooling and Nucleation

Phase transition is the process that changes the states of matter. In reality, phase transition is more than the simply thermodynamic picture we learned from undergraduate physical chemistry, which involved both heat and mass transfer at the interfaces between two or more phases. We are able to explain several important physical phenomena by applying our knowledge of fluid mechanics in Lecture 4 to phase transition. On example is the supercooling of liquid, that crystallization does not occur even if temperature is lower than the freezing point, due to the absence of nucleation. In this lecture we will study the phenomenon of nucleation during freezing, a process that nano- to microscale crystals called nuclei forms in the liquid phase. Nucleation is the first step of crystallization and also a kinetic process. Even when crystallization is thermodynamically favorable, nucleation can be slow or even unobservable under supercooling. What is the cause of supercooling? Can we engineer the nucleation process? We will find the answers in this lecture.

5.1 Thermodynamics of freezing

For a pure liquid with melting temperature T_m, the Gibbs free energy between its solid state (S) and liquid state (L) has the relation:

- $T > T_\mathrm{m} \rightarrow G_\mathrm{L} < G_\mathrm{S} \rightarrow$ Liquid is thermodynamically favored
- $T < T_\mathrm{m} \rightarrow G_\mathrm{L} > G_\mathrm{S} \rightarrow$ Solid is thermodynamically favored

The change of free energy G as a function of temperature T of a solid-liquid phase transition can be seen in Figure 5.1. The free energy of liquid phase G_L has a steeper slope than G_S, which results in a first-order phase transition (discontinuous $\partial G/\partial T$) around T_m.

The change of Gibbs free energy is: $\Delta G = G_\mathrm{S} - G_\mathrm{L} = \Delta H - T\Delta S$. The process L→S is exothermic and therefore $\Delta H < 0$. Since at $T = T_\mathrm{m}$, $\Delta G = 0$, we have: $\Delta S = \Delta H/T_\mathrm{m} < 0$. This makes sense since from liquid to solid the randomness of molecule orientation becomes smaller. If we assume that ΔH is independent of T, ΔG is related to the temperature perturbation ΔT near T_m:

$$\Delta G(T) = \frac{\Delta H \Delta T}{T_\mathrm{m}} \tag{5.1}$$

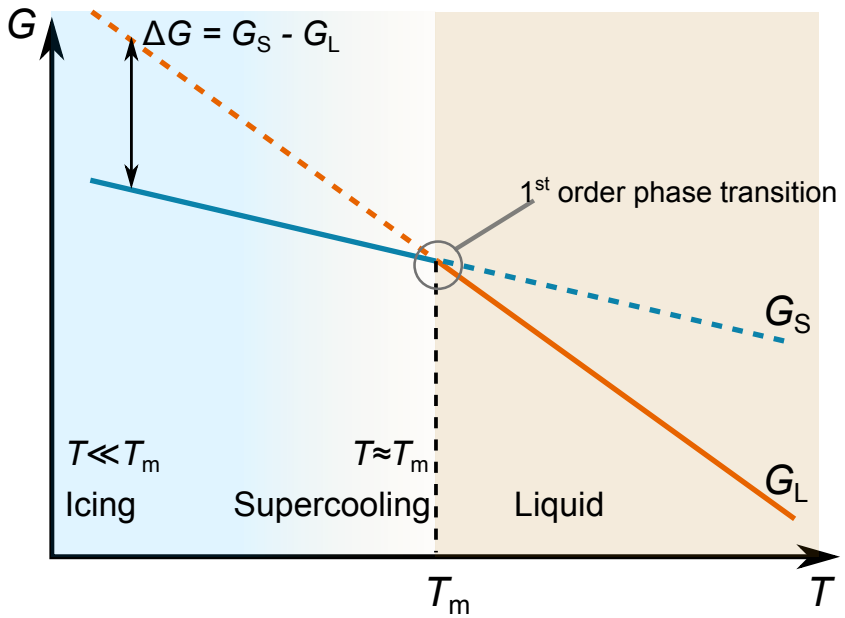


Figure 5.1: Free energy G of a solid-liquid phase transition. First order phase transition occurs at $T = T_{\mathrm{m}}$. The change of Gibbs free energy $\Delta G = G_{\mathrm{S}} - G_{\mathrm{L}}$ is a relative measure of the nucleation. When ΔG becomes more negative when T is much lower than T, and nucleation is more favored.

where $\Delta T = T_{\mathrm{m}} - T$. This means thermodynamically the process of crystallization is spontaneous once $T < T_{\mathrm{m}}$. However we usually observe that water does not freeze if we cool it down just below 0 °C. On the contrary, very pure liquid water can be kinetically stable until ∼-40 °C.[1,2] This phenomenon is known as the supercooling of liquid.

The crystallization process is a kinetic problem, since common sense indicates the freezing is much faster when ΔT = -40 °C, compared with ΔT = -0.1 °C. To model this we can view the nucleus as cluster of molecules. For a small molecule cluster A_n with n molecules of A, it can either form a larger cluster A_{n+1} through addition of one A molecule, or be formed from dissociating one A molecule from A_{n+1}:

$$A_{\mathrm{n}} + A \underset{k_d}{\overset{k_a}{\rightleftharpoons}} A_{\mathrm{n+1}}$$

where k_a and k_d are the kinetic constants for addition and dissociation, respectively. For the nucleation to occur one must have $k_a > k_d$. What is the cause for the disassociation if the free energy for phase transition is already negative? We will answer this in the next section.

5.2 Free energy of nucleation

There are two contributions to the free energy ΔG of nucleation: the volume part (ΔG_{V}) that comes from the free energy of phase transition, and the surface part (ΔG_{A}) that comes

from the creating of new interface

- Volume part ΔG_V (favorable)

  The volume part $\Delta G_V = \dfrac{\Delta H(T_m - T)}{T_m}$ is always smaller than 0, when $T < T_m$.

- Surface part ΔG_A (unfavorable)

  Phase transition increases the interfacial, which is energetically unfavorable ($\Delta G_A > 0$).

To combine both parts, we can calculate the total free energy change ΔG_n when the amount of molecules in the nucleus if n:

$$\begin{aligned} \Delta G_n &= \Delta G_V + \Delta G_A \\ &= n(G_S^m - G_L^m) + \eta n^{\frac{2}{3}} \gamma_{SL} \end{aligned} \tag{5.2}$$

where G_S^m and G_L^m are the molar free energies of solid and liquid phases, respectively, η is the shape factor which is defined as $\eta = (4\pi)^{\frac{1}{2}}(3\Omega)^{\frac{2}{3}}$ where Ω is the molecular volume, assuming spherical nucleus.[3] As shown in Figure 5.2, since $\Delta G_{V,n} \sim -n$ and $\Delta G_{A,n} \sim n^{\frac{2}{3}}$, eventually the volume part overcomes the surface part when the cluster is large enough. When n is larger than the critical cluster size n_c, $\partial \Delta G_n / \partial n < 0$ and the clusters continue to grow. The value for n_c can be determined by solving $\partial \Delta G_n / \partial n = 0$ using Equation 5.2,

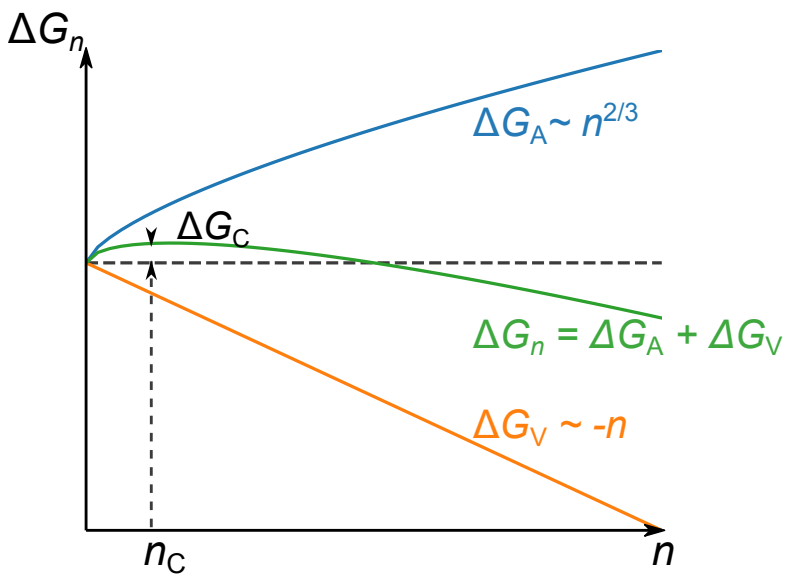


Figure 5.2: Volume and surface contributions to the free energy ΔG_n during nucleation, as a function of the cluster size n. A energy barrier of ΔG_c at $n = n_c$ exists.

which gives:

$$n_c = -\frac{8}{27}\left[\frac{\eta \gamma_{SL}}{G_S^m - G_L^m}\right]^3 \tag{5.3}$$

When $n = n_c$, there is a potential barrier ΔG_c that prevents the nucleation:

$$\Delta G_c = \frac{1}{3}\eta\gamma_{SL}n_c^{\frac{2}{3}} = \frac{4}{27}\eta^3\gamma_{SL}^3\frac{T_m^2}{(\Delta H\Delta T)^2} \tag{5.4}$$

Look again at Figure 5.1, when T is only slightly smaller than T_m, ΔG_V is not enough to overcome the surface energy ΔG_A. From Equation 5.4, the energy barrier $\Delta G_c \propto \frac{\gamma_{SL}^3}{(\Delta T)^2}$. For small ΔT below T_m, the potential barrier is high enough to keep the liquid phase kinetically stable. From classical nucleation theory (CNT), the probability to form a cluster with size n_c, p_c, is:[3] $p_c = N_{tot}\exp\left(\frac{-\Delta G_c}{kT}\right)$, where N_{tot} is the total number of nuclei. When ΔG_c decreases, the probability of forming a nucleus with size n_c increases. The rate of nucleation $\dot{N}$ (how many new nuclei forms per unit time) can be estimated using:[3]

$$\dot{N} \sim kp_cZ \tag{5.5}$$

where k the rate for a molecule in liquid to join a solid cluster and Z is the Zeldovich imbalance factor that considers the dissociation of nuclei.[4] The rate k_n is associated with the total number of nuclei n_{tot}, the diffusivity $\mathcal{D}_{AB}$ and the critical surface area A_c of a cluster with $n = n_c$:

$$k = \frac{3}{4}N_{tot}^{4/3}\mathcal{D}_{AB}A_c \tag{5.6}$$

On the other hand, Z is associated with ΔG_c, n_c, and T:

$$Z = \sqrt{\frac{\Delta G_c}{3\pi n_c^2 k_B T}} \tag{5.7}$$

Plug in all the parts into Equation 5.5 we have the expression for $\dot{N}$:

$$\dot{N} = \frac{3}{4}N_{tot}^{\frac{7}{3}}A_c\mathcal{D}_{AB}\left(\frac{\Delta G_c}{3\pi n_c^2 k_B T}\right)^{\frac{1}{2}}\exp\left(-\frac{\Delta G_c}{k_B T}\right) \tag{5.8}$$

which tells us the nucleation rate can be controlled by $\mathcal{D}_{AB}$, T, ΔG_c, etc. Note that these variables are not completely independent (e.g. both $\mathcal{D}_{AB}$ and ΔG_c depend on T).

5.3 Crystallization of solute

The above analysis for nucleation in pure liquid can also be applied to a solution of molecule A. The only difference is that ΔG of crystallization becomes the difference of chemical potentials μ_A in solid and liquid phases:

$$\Delta G = \mu_A^S - \mu_A^L \tag{5.9}$$

As seen from the solubility diagram of the solution system (Figure 5.3), at temperature T, a saturated solution of A has concentration $c^\star(T)$. Crystalline can be achieved by either

reducing the temperature or increasing concentration. Let's consider the analog to supercooling, that crystallization is induced by decreasing solution temperature T at the same concentration. There is a metastable region when the solute concentration c is slightly smaller than $c^\star(T)$, where nucleation rate is slow, similar to the case of supercooling. The

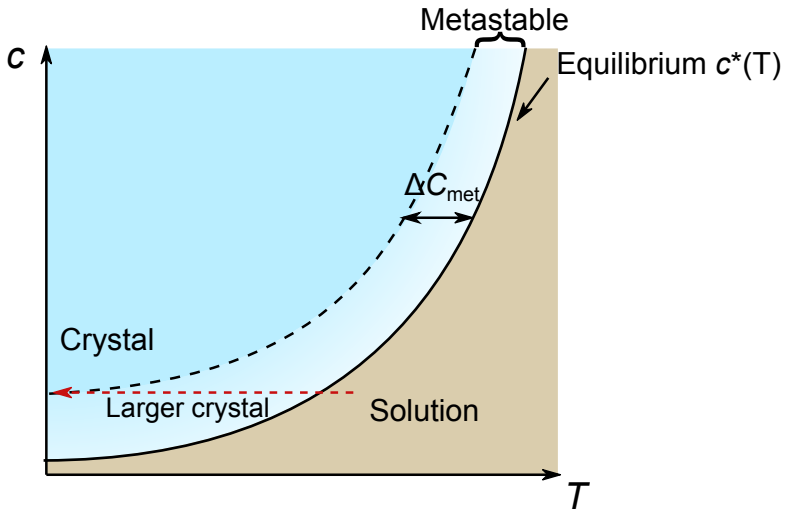


Figure 5.3: Typical solubility diagram of concentration c as a function of temperature T. The equilibrium solubility is $c^\star$. When c is slightly smaller than $c_\star$, there is a metastable region where nucleation rate is slow, analog to supercooling.

width Δc_{met} of the metastable region depends on both $\dot{N}$ and c^*. A practical estimation is $\Delta c_{\text{met}} \propto 1/\dot{N}$.[5,6] In industry, larger crystals can be grown with a larger metastable width Δc_{met}. Therefore it is desirable to have smaller $\dot{N}$. Using Equation 5.8, this can be achieve by:

- Reducing T
- Reducing $\mathcal{D}_{\text{AB}}$

From the Einstein-Stokes equation $\mathcal{D}_{\text{AB}} = kT/(6\pi\eta r)$ (will be discussed in Lecture 15), where η is the viscosity of the liquid and r is the radius of the soluble molecule. Reducing T already decreases $\mathcal{D}_{\text{AB}}$. We can also use a solvent with larger viscosity to achieve a smaller $\mathcal{D}_{\text{AB}}$.

5.4 Heterogeneous nucleation

The above discussions are based on the assumption that nucleation forms in the bulk solution. However in experiments, we usually find the nucleation is more preferred on the wall of container (heterogeneous nucleation) instead of in the bulk solution (homogeneous nucleation). We can understand the mechanism of heterogeneous nucleation by studying the volume and surface contributions to the free energy during nucleation. If the nuclei are

small enough, we can simplify them as "droplets". On the wall of container, such "nucleus droplet" (n) has contact angle θ, and a spherical shape with radius R (Figure 5.4).

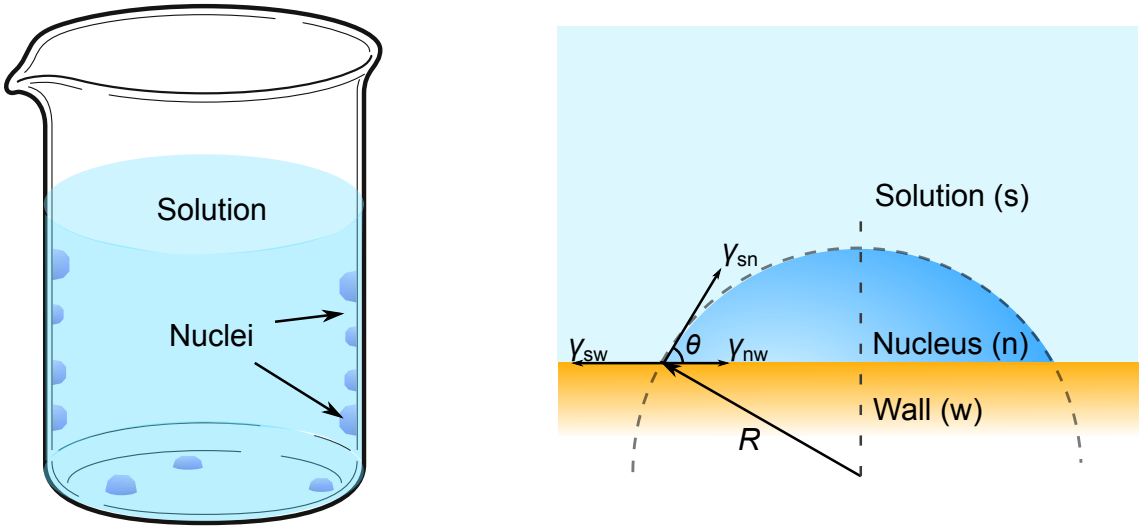


Figure 5.4: Heterogeneous nucleation. Left: heterogeneous nucleation occurs on the wall of container. Right: simplified geometry of a nucleus on wall. The nucleus is assumed to be a spherical droplet with contact angle θ.

The geometry of the droplet gives:

- Volume of nucleus: $V_n = \frac{\pi R^3}{3}(2 - 3\cos\theta + \cos^3\theta)$
- Interfacial area of nucleus with solution: $A_s = 2\pi R^2(1 - \cos\theta)$
- Interfacial area of nucleus with wall: $A_c = \pi R^2 \sin^2\theta$

The total free energy of the system considering volume and surface contributions is:

$$\Delta G = V_n \Delta G_V + \gamma_{sn} A_s + (\gamma_{nw} - \gamma_{sw}) A_c \tag{5.10}$$

where the index s, n and w denote the solution, nucleus and wall, respectively. The interfacial tensions are linked by the Young equation:

$$\gamma_{sw} = \gamma_{nw} + \gamma_{sn}\cos\theta \tag{5.11}$$

The heterogeneous nucleation free energy is then:

$$\Delta G_{het} = \underbrace{\left[\frac{4\pi R^3}{3}\Delta G_v + 4\pi R^2 \gamma_{sn}\right]}_{\Delta G_{homo}} \left[\frac{2 - 3\cos\theta + \cos^3\theta}{4}\right] \tag{5.12}$$

We can see that the heterogeneous and homogeneous nucleation free energy differ only by a factor:

$$\frac{\Delta G_{het}}{\Delta G_{homo}} = \frac{2 - 3\cos\theta + \cos^3\theta}{4} = f \tag{5.13}$$

As seen in Figure 5.5a, the value of f monotonically increases with θ. As a consequence, the heterogeneous nucleation free energy ΔG_{het} decreases when θ is smaller. If $\theta \to 0$, the heterogeneous nucleation barrier $\Delta G_{\text{c,het}} \to 0$, the nucleus completely wets the wall and the heterogeneous nucleation is spontaneous. For θ form 0° to 180°, $\Delta G_{\text{het}} \leq \Delta G_{\text{homo}}$, which means heterogeneous nucleation is always preferred over homogeneous nucleation, since ΔG_{c} for heterogeneous nucleation is lowered due to larger nucleus-wall interaction (Figure 5.5b). To suppress the heterogeneous nucleation and get large and uniform crystals

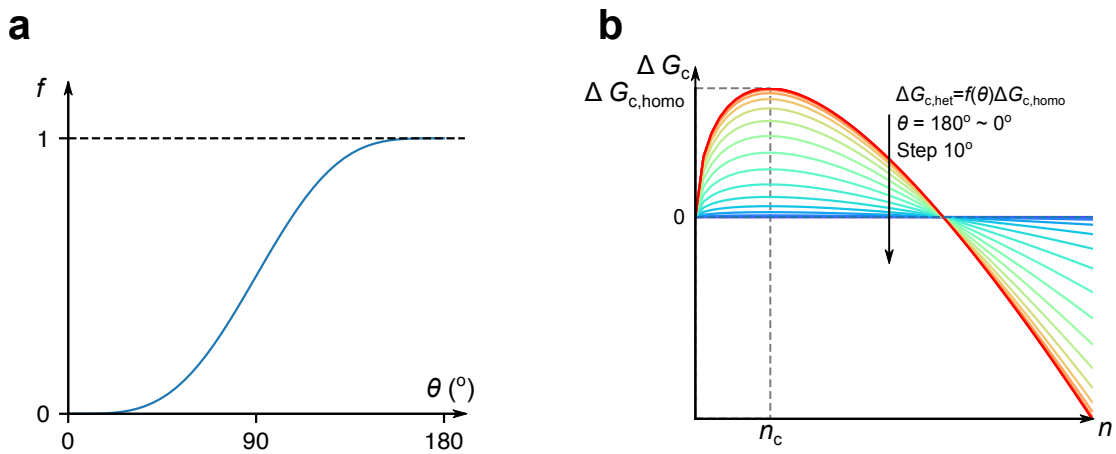


Figure 5.5: (a) Value of f (ratio between ΔG_{het} and ΔG_{homo}) as a function of contact angle θ. (b) Evolution of ΔG_{het} with θ. When θ drops the nucleation barrier decreases.

grown from bulk solution, we need to find a low-energy wall with larger contact angle θ, such as teflon. Since $\dot{N} \sim \exp(-\frac{\Delta G_{\text{c}}}{k_{\text{B}}T})$, increasing θ also suppress the nucleation rate. On the other hand, we can also enhance the heterogeneous nucleation, by reducing the surface contact angle. In this case rough (fractal) surfaces are usually involved due to the Wenzel-like wetting state[7] which we discussed in Lecture 2.